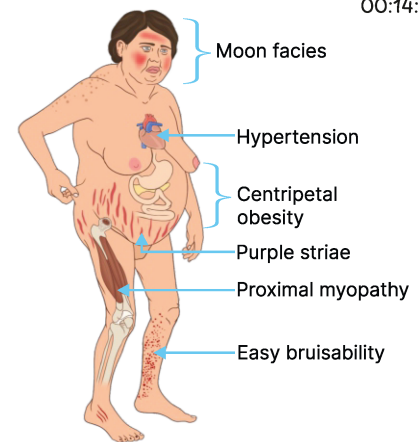


## HYPERCORTISOLISM - CLINICAL FEATURES

- Moon facies : Facial plethora
- HTN : Diastolic
- Buffalo hump / dorso - cervical fat pad
- Centripetal Obesity
- Purple striae
- Proximal myopathy : Ex - difficulty in climbing stairs
- Easy bruisability

**Hypertension with hypokalemic acidosis is seen in Conn's syndrome**



00:14:07

## OTHER FEATURES SEEN IN CUSHING'S SYNDROME

- ↑↑ blood sugar
  - Cortisol ↑ : Sugar ↑
  - Weight gain : Insulin resistance → Sugar ↑ (FBS > 126mg/dl)
- HTN with hypokalemic alkalosis
  - Cortisol ↑ : ENac ++ → ↑ salt and H<sub>2</sub>O → More loss of K<sup>+</sup>/H<sup>+</sup>
- Hirsutism, Amenorrhea, Weight gain
  - Cortisol ↑ : Sex steroids receptors(+) → Androstenedione effects
- Hyperpigmentation of knuckles and surgical scars +/-
  - Seen with ectopic ACTH production : Carcinoid tumor
- Hypercoagulable state - Increased incidence of DVT/Pulmonary embolism
- Eosinopenia & ↑ susceptibility to infections

00:15:12

## MNEMONIC FOR FEATURES OF CUSHING SYNDROME : COP-HAS-ID

- Central adiposity
- Osteoporosis
- Proximal myopathy
- Hirsutism
- Amenorrhea
- Striae purplish, > 1 cm, Thinning of the skin
- Impaired glucose tolerance
- Diastolic hypertension

## WORK UP

00:25:40

## SCREENING TEST AND CONFIRMATION OF DIAGNOSIS

- 24-hour urinary cortisol: ↑↑ → ≥2x
- Dexamethasone Overnight Test (DOT):
  - 1 mg tablet of Dexamethasone at 11 pm, plasma cortisol at 8 am > 50 nmol/L
- IOC - Midnight Salivary Cortisol ↑ : > 5 nmol/L
- If still needed for confirmation, then perform Low dose dexamethasone test (0.5 mg dexamethasone 6 hourly for 2 days, plasma cortisol > 50 nmol/L)
- Plasma ACTH levels



| > 15 pg/ml                                                                                                                                                                                                                                                                                                                                                                                                                             | < 5pg/ml                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>ACTH-dependent Cushing syndrome</p> <p>↓</p> <ul style="list-style-type: none"> <li>• MRI pituitary</li> <li>• CRH test → CRH 100 microgram IV, ACTH↑ &gt;40% at 15 min</li> <li>• High-dose dexamethasone suppression test(Differentiate B/w lung ca/pituitary adenoma)               <ul style="list-style-type: none"> <li>◦ Cortisol suppression &gt; 50% after 2 mg Dexamethasone given 6 hrly X 2 days</li> </ul> </li> </ul> | <p>ACTH-independent Cushing syndrome</p> <p>↓</p> <p>Unenhanced CT abdomen</p> <p>↓</p> <p>Adrenal Adenoma</p> <p>↓</p> <p>U/L or B/L adrenalectomy</p> |

### SCENARIOS POSSIBLE WITH BELOW MENTIONED TEST

00:36:02

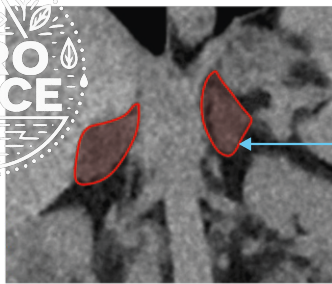
| CRH +VE (40% spike),HIGH DOSE DEXAMETHASONE TEST +VE( 50% suppression) | CRH & HIGH DOSE DEXAMETHASONE TEST EQUIVOCAL (+/-)                                                                                                                                                          | CRH -VE,HIGH DOSE DEXAMETHASONE TEST -VE                                                                                                     |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Impression - Cushing disease                                           | <ul style="list-style-type: none"> <li>• Test done - Inferior petrosal sinus sampling</li> <li>• If sinus/peripheral ACTH ratio is &gt; 2 → pituitary source (Pituitary adenoma/Cushing disease)</li> </ul> | <ul style="list-style-type: none"> <li>• Impression - Ectopic source</li> <li>• MRI Abdomen/CT is done to find the ectopic source</li> </ul> |
| T/t plan - Trans-sphenoidal surgery                                    | T/t - Trans-sphenoidal surgery                                                                                                                                                                              |                                                                                                                                              |

| ACTH INDEPENDENT CUSHING SYNDROME         | ACTH-DEPENDENT CUSHING SYNDROME                                                                                                                                                                                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adrenal adenoma: U/L or B/L adrenalectomy | Cushing disease - Trans-sphenoidal surgery                                                                                                                                                                                                                             |
| Adrenocortical carcinoma - Mitotane       | <p><b>An ectopic source with metastasis</b></p> <ul style="list-style-type: none"> <li>• Metapyrone: 11β hydroxylase (oral)</li> <li>• Ketoconazole (oral)</li> <li>• Osilodrostat - Inhibit aldosterone synthase (oral)</li> <li>• Etomidate infusion (IV)</li> </ul> |

### NELSON SYNDROME

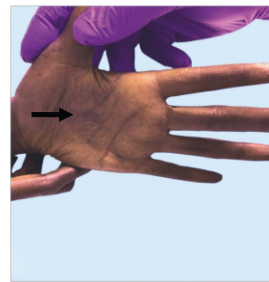
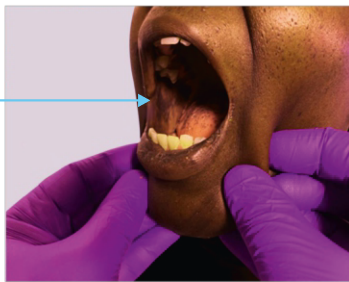
00:44:43

- Cause: **B/L adrenalectomy**
  - Loss of inhibitory control over ACTH release
  - ACTH ↑↑ (MSH-like action)
- Intra-op: IV Hydrocortisone : prevent Addisonian crisis
- Post-op: Oral Hydrocortisone on a lifelong basis
- Manifestation : Follow-up visits → Hyperpigmentation over knuckles/creases/oral mucosa



Hyperpigmentation

B/L Adrenal  
adenoma



### MCQ

Q. Which investigation should be first done for diagnosis of ACTH-dependent Cushing's syndrome?

- a. CRH test
- b. High-dose dexamethasone test
- c. Inferior petrosal sinus sampling
- d. MRI head

Ans: (d)

00:49:15



# DISEASES OF THYROID

## THYROID STORM

00:01:00

- Medical emergency with a high level of mortality

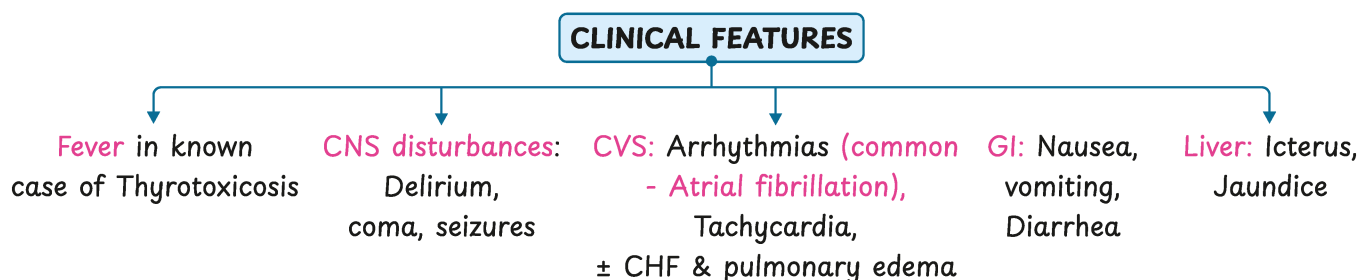
## TRIGGERS

00:01:10

- **Acute illness:** Stroke, infection like Covid-19, DKA, trauma
- **Surgery** (Thyroid surgery of patient with Hyperthyroidism)
- **Radioiodine** to partially treated patients of Hyperthyroidism → destruction of overactive gland → Massive release of thyroid hormone into circulation → extensive Calorigenesis ( $\uparrow T_3$ ,  $\uparrow T_4$ ) & heat production

## CLINICAL FEATURES

00:03:53



- Causes of tachycardia: Intrinsic sympathomimetic properties of thyroid hormones, heat production in body.

## BURCH WARTOFSKY SCALE - USED TO CONFIRM DIAGNOSIS OF THYROID STORM

- If the **value is > 45** → **Confirm diagnosis of thyroid storm**

### Point Scale for the Diagnosis of Thyroid Storm

Thermoregulatory dysfunction  
Temperature (8°F)

|             |    |
|-------------|----|
| 99.0-99.95  | 0  |
| 100.0-100.9 | 10 |
| 101.0-101.9 | 15 |
| 102.0-102.9 | 20 |
| 103.0-103.9 | 25 |
| >104.0      | 30 |

Gastrointestinal-hepatic dysfunction  
Manifestation  
Absent 0

Moderate (diarrhea, abdominal pain, nausea/vomiting) 10  
Severe (jaundice) 20

### Cardiovascular

Tachycardia (beats per minute)

|         |    |
|---------|----|
| 100-109 | 5  |
| 110-119 | 10 |
| 120-129 | 15 |
| 130-139 | 20 |
| >140    | 30 |

### Central nervous system disturbance

Manifestation

Absent 0

Mild (agitation) 10

Moderate (delirium, psychosis, extreme lethargy) 20

Severe (seizure, coma) 30

### Atrial fibrillation

|                          |    |
|--------------------------|----|
| Absent                   | 0  |
| Present                  | 10 |
| Congestive heart failure |    |
| Absent                   | 0  |
| Mild                     | 5  |
| Moderate                 | 10 |
| Severe                   | 20 |

### Precipitant history

Status

Positive 0

Negative 10

Scores totaled

>45 **Thyroid storm**

25-44 **Impending storm**

<25 **Storm unlikely**

## TREATMENT

00:06:55

- Cooling - Ice packs can be used
- Oxygenation
- IV Fluids - To compensate for dehydration
  - In case of CHF features → IVF is stopped
- **DOC** - **Propylthiouracil** (PTU)
  - **MOA** - ↓ production of T4  
→ **Also inhibits peripheral deiodination from T4 to T3**
  - **ROA** - NG Tube, rectal route
  - **Alternative drug** - Methimazole
- Saturated solution of Potassium iodide on a 6 hourly basis → Work via **Wolf Chaikoff effect**
  - Give Mega doses of Iodine (6 hourly basis) → ↓ Iodine trapping → ↓ **production of T4 & T3**
- **Propranolol** - Blocks  $\beta_1$  &  $\beta_2$  → Counterbalance sympathomimetic stimulation blocks the tachycardia component and muscle effects
- **Esmolol** - Recommended in tachycardia with hypotension
- **Hydrocortisone**
- **Cholestyramine** → Causes Sequestration of thyroid hormones

### Important Information

- Objective - To ↓ total amount available biologically active thyroid hormone i.e., T4 in system



### JOD BASEDOW EFFECT

Start with an intake of iodized salt (**minute dose of iodine**) for a long duration ( $I_2$  replete population)

↓  
↑ iodine trapping → ↑ T4 production

↓  
Thyrotoxicosis

### WOLFF CHAIKOFF EFFECT

Saturated solution of potassium iodide (**Mega doses of iodine** are given on 6 hourly basis)

↓  
↓ Iodine trapping

↓  
↓ T4 & T3

### THYROTOXICOSIS FACTITIA

00:17:45

- Intake of cattle/beef meat containing the thyroid gland of an animal
- Intake of illegal medicine ordered over the internet for weight loss (generally contain Levothyroxine extract)
- **Exogenous intake of T4** in both these situations → ↑ in Basal metabolic rate, Weight loss, HR, Angina, Tremors (**all features of thyrotoxicosis**)

### HYPERTHYROIDISM & OTHER CAUSES OF THYROTOXICOSIS

00:21:55

- **Hyperthyroidism** - ↑ function of the thyroid gland
- **Thyrotoxicosis: Excess T4 levels in blood** (d/t exogenous intake of thyroid hormone, iodized salt)

### CAUSES OF THYROTOXICOSIS WITHOUT HYPERTHYROIDISM

00:23:37

- Subacute thyroiditis (Inflammatory destruction of thyroid gland follicles)
- Silent thyroiditis (Destruction of thyroid gland over a period of time)
- Radiation, Infarction of thyroid tumor (adenoma)
- Thyrotoxicosis **factitial**
- Amiodarone (Drug induced hyperthyroidism)

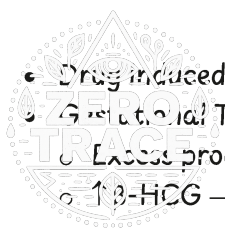
### CAUSES OF HYPERTHYROIDISM

00:27:28

| PRIMARY HYPERTHYROIDISM                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SECONDARY HYPERTHYROIDISM                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• M/c cause → GRAVE'S DISEASE: Autoimmune disorder common in young female → have <b>TRAb</b> (TSH Receptor antibody)/ LATS (Long-acting Thyroid Stimulating Ab)</li> </ul> <p>TRAb acts on the TSH receptor present on thyroid follicles</p> <p>↓</p> <p>T4 ↑, T3 ↑</p> <p>↓ Feedback mechanism</p> <p>↓ <b>TSH</b> (sometimes undetectable)</p> <ul style="list-style-type: none"> <li>• Next Best examination - Check for TRAb</li> </ul> | <ul style="list-style-type: none"> <li>• TSH producing Pituitary adenoma: ↑ <b>TSH</b> → <b>T4</b> ↑ &amp; <b>T3</b> ↑</li> <li>• Next Best examination - MRI</li> </ul> |

### OTHER CAUSES OF HYPERTHYROIDISM

- Toxic Multinodular Goitre/ Toxic adenoma/ activation mutation of TSH receptor - primary cause
- Struma Ovarii - Ovarian tumor causing production of thyroid hormone
  - Primary cause



- Drug induced (Jod Basedow effect) - Primary cause
- Gestational Trophoblastic Neoplasia - Secondary cause
- Excess production of  $\beta$ -HCG (chemical structure slightly similar to TSH)
- $\uparrow \beta$ -HCG  $\rightarrow \uparrow$  **TSH**  $\rightarrow$  **T4**  $\uparrow$  & **T3**  $\uparrow$

## CLINICAL FEATURES

00:34:58

- Hyperactivity, irritability, Dysphoria  $\rightarrow$  **Earliest manifestation**
- In young female
  - Palpitations  $\rightarrow$  d/t  $\beta_1$  stimulation
  - **Fine tremors (characteristic), Sweaty palms**  $\rightarrow$  d/t  $\beta_2$  stimulation
  - Sympathomimetic stimulation on  $\alpha_1 \rightarrow$  vasoconstriction  $\rightarrow$  Hypertension, Headache, tiredness
  - **BMR  $\uparrow \rightarrow$  Weight loss**
  - **$\uparrow$  Calorigenesis  $\rightarrow$  Heat intolerance**
  - Oligomenorrhea
  - Infertility
  - **Proptosis**

## EXAMINATION FINDINGS

00:38:11

- **Sleeping Pulse Rate** (done to differentiate from anxiety neurosis)  $\rightarrow$  increased, resting tachycardia
- Arrhythmias: Atrial fibrillation (common) - irregularly irregular pulse
- Loud S1/ Ejection systolic murmur } d/t Hyperdynamic precordium/
- Means Lerman Scratch } circulation
- **Pretibial Myxedema** - Orange discoloration around shin: Specific for **grave's disease**
  - **Lesions on shin:** Pyoderma gangrenosum, Necrobiosis Lipidoica Diabetiarum (more specific for type I DM)
- **Grave's ophthalmopathy**  $\rightarrow$  Proptosis
- **Acropachy**

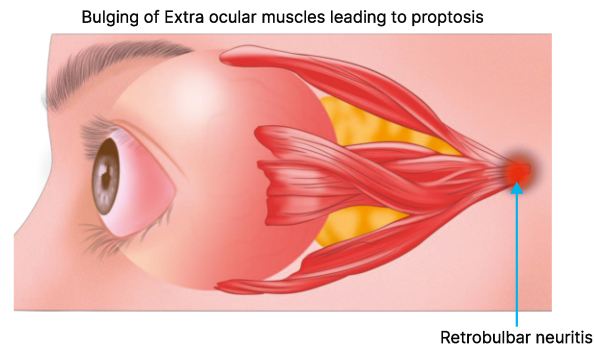


## GRAVE'S OPHTHALMOPATHY

00:44:42

- TSH Ab also cause damage to muscles  $\rightarrow$  Swelling of eye muscles
- Swelling of eye muscles + retrobulbar fat deposition  $\rightarrow$  push eyeball anteriorly
- Earliest sign - **Stare sign**
  - Muscle involved: Muller's muscle (muscle of upper eyelid) supplied by ciliary ganglion
  - $\uparrow$  T4 & T3 (intrinsic sympathomimetic drive)  $\rightarrow$  activation of ciliary ganglion  $\rightarrow$  **contraction of mullers muscle**  $\rightarrow$  **retraction of upper eyelid**
- Lid lag sign
- Periorbital edema
- Extraocular muscle involvement: M/c - **Inferior rectus** causing inferior rectus palsy  $\rightarrow$  **Diplopia**
- Exposure keratitis
- **Sight loss:** d/t damage to cornea
- Retrobulbar neuritis





| NO SPECS SCHEME |                                                 |
|-----------------|-------------------------------------------------|
| 0               | No signs or symptoms                            |
| 1               | Only signs (lid retraction or lag), no symptoms |
| 2               | Soft tissue involvement (periorbital edema)     |
| 3               | Proptosis (>22 mm)                              |
| 4               | Extraocular muscle involvement (diplopia)       |
| 5               | Corneal involvement                             |
| 6               | Sight loss                                      |

### TREATMENT

- Antithyroid drugs
- For inflammatory damage or to prevent possibility of blindness - **Methylprednisolone**

### PEMBERTON SIGN

- **Retrosternal extension of enlarged thyroid**
- **Facial congestion/ Facial Plethora** on arm elevation due to inferior displacement of enlarged thyroid gland pressing on superior vena cava

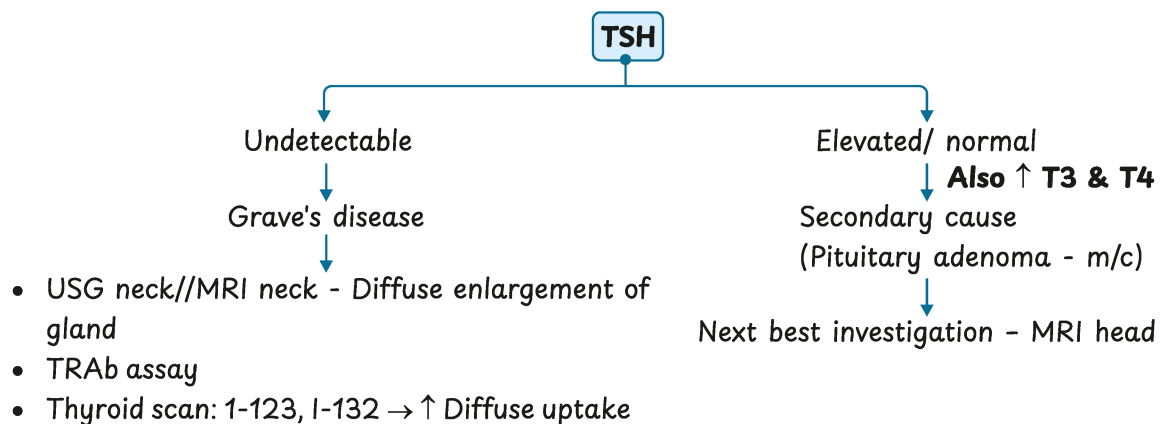


00:53:42

### WORK UP

- Thyroid Function Tests: Total  $\uparrow$  T4 & T3  $\uparrow$ , Free T4 & T3  $\uparrow$
- **For diagnostic purposes: Free/unbound T4 & TSH**

00:55:54





### Important Information

**Symptomatic hyperthyroidism** - Normal levels of Free T4 and T3 and ↓ TSH

### TREATMENT

01:02:39

- Lab report should be ↓ TSH, ↑ T4 & T3
- Grave's disease: Lab confirmed thyrotoxicosis with diffuse enlargement of thyroid with eye signs.
- Additional test (in case all 3 above mentioned features are not present): USG, TRAb, Thyroid scan
- TOC: According to American thyroid association - **Radioactive iodine/Thyroid ablation**
  - Form of iodine for thyroid ablation - **Iodine 131**
    - $t_{1/2} = 8$  days, ablation/ destruction of thyroid gland with Beta rays
- **First line treatment** - Antithyroid drugs: Propylthiouracil & Carbimazole
  - To achieve euthyroid status
  - **DOC - Carbimazole**
  - Propylthiouracil has high protein binding: Good for female planning to conceive or are in 1<sup>st</sup> trimester.

### ANTITHYROID DRUGS

| PTU (Propylthiouracil)                                                                                                                                               | CARBIMAZOLE                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Safe in pregnancy (first trimester T1).</li><li>• <b>Black box warning:</b> Hepatotoxic potential on prolonged use</li></ul> | <ul style="list-style-type: none"><li>• <b>Safe in 2<sup>nd</sup> Trimester</b></li><li>• Teratogenic effect: Aplasia cutis and Choanal atresia</li></ul> |

### Important Information

- Dose-limiting side effect of Antithyroid drugs - Agranulocytosis

### TREATMENT OF 2° HYPERTHYROIDISM

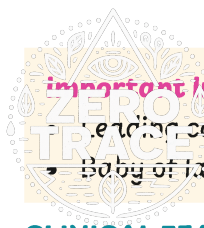
01:12:06

- If Lab report - ↑ TSH, ↑ T4 & T3 → TSH secreting pituitary adenoma
- T/t - Initially antithyroid drugs
- **TOC: Transsphenoidal Surgery**

### HYPOTHYROIDISM

01:13:51

| PRIMARY HYPOTHYROIDISM                                                                                                                                                                                                                                                                                                                            | SECONDARY HYPOTHYROIDISM                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Thyroid gland involvement</li><li>• T4 ↓, T3 ↓, <b>TSH ↑</b></li></ul>                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>• Pituitary gland damage</li><li>• T4 ↓, T3 ↓, TSH ↓</li></ul>                                                                                                                                                                                                 |
| <b>CAUSES</b> <ul style="list-style-type: none"><li>• Iodine deficiency disorder/<b>Endemic cretinism</b></li><li>• Hashimoto thyroiditis: TPO Ab (Thyroid peroxidase Ab) damaging thyroid follicles</li><li>• Food Goitrogen → Consumption of Cabbage, Turnip in large amounts → ↓ synthesis of T3</li><li>• Drugs - <b>Amiodarone</b></li></ul> | <b>CAUSES</b> <ul style="list-style-type: none"><li>• Sheehan syndrome</li><li>• <b>Simmonds's disease</b> (Arteriovenous malformation, Tumor, Hypertension) - <b>Non obstetric cause of damage to pituitary</b></li><li>• CNS irradiation</li><li>• Sarcoidosis</li><li>• Hemochromatosis</li></ul> |



### Important Information

- Leading cause of Congenital Hypothyroidism - Thyroid dysgenesis
- Baby of lady living in area of iodine deficiency is born with hypothyroidism - **Endemic cretinism**

### CLINICAL FEATURES

01:19:59

- Reduced BMR - Weight gain (modest in comparison to Cushing syndrome)
- ↓ Calorigenesis - **Cold intolerance**
- Myxedema**
  - Scalp: Thinning of hair, Alopecia, thinning of eyebrow hair (Queen Annie sign)
  - Skin: Dry, Coarse skin
  - Face: Puffy face
  - GIT: **Constipation** → Poor response to laxatives
  - Vocal cords affected - Hoarseness of voice
- Menstrual issues: Menorrhagia, Infertility
- CNS manifestations: **Hashimoto encephalopathy**
  - Myoclonus, EEG: slow waves, **steroid responsiveness**

### REASON FOR DIARRHOEA IN HYPERTHYROIDISM & CONSTIPATION IN HYPOTHYROIDISM

01:24:58

- Rationale **for diarrhea in thyrotoxicosis**
  - Increased appetite → ↑ Carbohydrate intake → ↑ **Osmotic load** in gut
  - Sympathomimetic stimulation due to thyroid hormone → ↓ **gut motility** → Bacterial count ↑ → **Osmotic diarrhea**
  - Altered bile acid metabolism.
- Rationale for constipation
  - Less calorie intake
  - Less bacterial count
  - Less thyroid hormone

### EXAMINATION FINDINGS

01:27:20

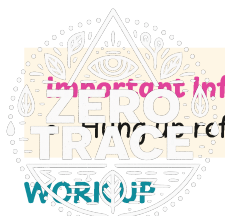
- Hung up ankle jerk/ Woltman sign** - Myxedema of Achilles tendon cause slow/ delayed relaxation of ankle jerk → **Most reliable sign**
- Isolated diastolic hypertension: BP=120/100mmHg
  - ↑ peripheral resistance in body due to weight gain → HTN
- Heart rate < 60/min
- Thinning of eyebrow hair (Queen Annie sign) - Early feature but not specific
- β-Carotenemia → Lemon yellow tint of the skin
  - β-Carotene → ~~X~~ → Vitamin A

T4



Hung up Ankle jerk

- In iodine deficiency areas → ↓ T3 & T4 → By feedback mechanism ↑ TSH & also ↑ TRH → ↑ prolactin → Stimulate acini of breast causing **Galactorrhea**
  - Seen in
    - **Primary hypothyroidism**
    - Prolactinoma
    - Chronic renal failure



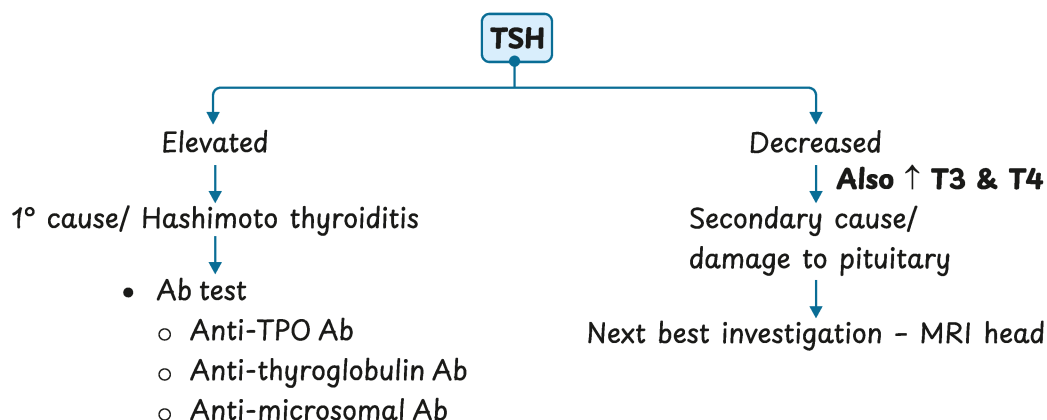
## Important Information

- Tendon reflexes - seen in chorea

01:36:06

### Thyroid function test

- Total T4, T3- ↓
- Free/ unbound T4, T3- ↓
- Preferred/ limited test - free T4, TSH



### USG/ MRI neck

- USG/ MRI of neck - Goitre (can be retrosternal), enlargement of thyroid gland

## ADDITIONAL INVESTIGATIONS

- ECG
  - Serous cavity effusions → Myxedema heart (Lymphedema outside the heart)
  - Low voltage ECG → Massive pericardial effusion (slow, gradual accumulation of fluid around heart that chamber of heart adjusts according to that gradual increase) → **Pulsus paradoxus absent**
- Chest x-ray - Apparent increase in cardiac-thoracic ratio or **Water bottle appearance** → Myxedema heart
- CBC - Macrocytic anemia
  - Causes of Macrocytic anemia
    - B12 deficiency
    - Folic acid deficiency
    - **Aplastic anemia**
    - **Chronic liver disease**
    - **Hypothyroidism (T3 ↓, T4 ↓)**
- Lipid profile: ↑ Cholesterol
  - Accelerated atherosclerosis

## UNIVERSAL TREATMENT

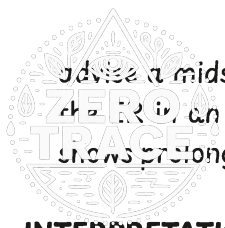
01:44:34

- Levothyroxine - Starting dose: 1.6 µg/kg/day (lifelong)
- In older patients > **65 years** with **coexistent CAD** → Start with lower doses i.e., 25µg OD & increase dose gradually to prevent worsening of cardiac ischemia

## CASE SCENARIO

01:47:45

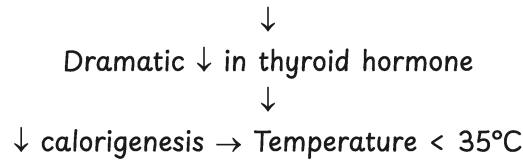
- A 60-year-old lady who is diagnosed with hypothyroidism and insomnia develops symptoms of UTI. You



advise a midstream urine culture followed by antibiotics. After 2 days of consultation, she was brought to the ER in an unconscious state. Vitals are pulse 58/min, BP-100/60 mmHg, and temperature 35°C. ECG shows prolonged PR interval and patient remains hypothermic despite the use of blankets.

## INTERPRETATION

- **Diagnosis - Myxedema Coma**
- UTI/ Sleeping pills (d/t insomnia) - Trigger in patient of hypothyroidism whose non-compliant/ partially compliant



- Treatment
  - Rewarming of the body
  - Liothyronine (T3 preparations), Levothyroxine

## SICK EUTHYROID SYNDROME

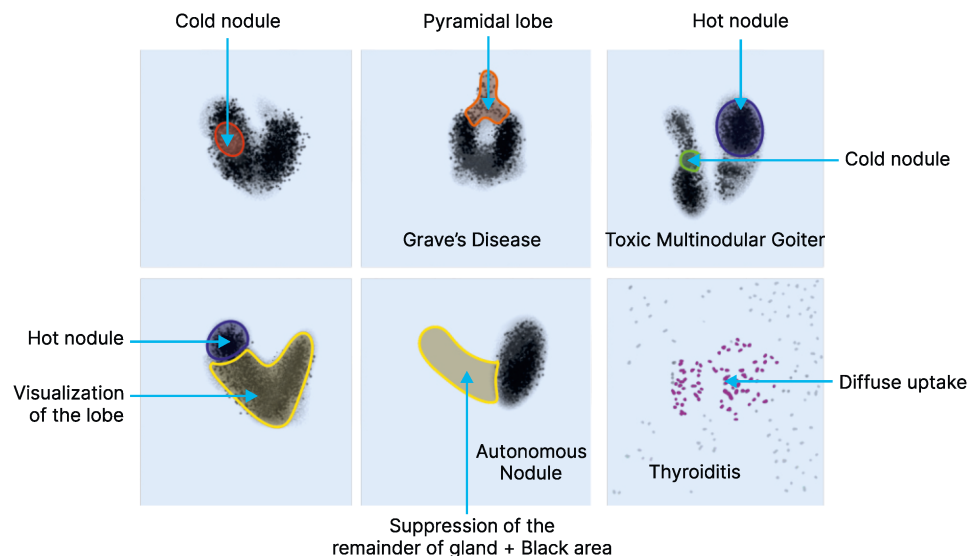
01:52:07

- Trigger - Sepsis
- **Impaired conversion of T4 to T3**
- **T4 - normal, T3 ↓**
- **TSH - ↓ initially subsequently with treatment TSH → normal/↑ (recovery phase)**
- T/t - Treat the cause

## THYROID SCAN

01:53:48

- Radioactive iodine used - I 132 & I 123
- Half life
  - I-132 - 13.2 hours
  - I-123 - 2.3 hours
  - I-131 - 8 days
- Surgery needs to be done in Toxic multinodular goitre





- Q. A 35-year-old female presents with symptoms of feeling anxious all the time with weight loss, tremors, and palpitations. She has diffused enlargement of the thyroid with loud S1 and bounding pulses. Which is correct about the possible etiology of presentation?
- a. Autoimmune, Activation of TSH-receptors
  - b. Autoimmune, inactivation of TSH-receptors
  - c. Inadequate conversion of T4 to T3 in peripheral tissues
  - d. Excess production of TSH by anterior pituitary

Ans: (a)

- Q. A 65-year-old female patient presents to the clinic with complaints of fatigue, weight gain, and cold intolerance. On examination, her blood pressure is 130/80mmHg, heart rate is 60 bpm, and weight is 50 kg. Laboratory tests reveal elevated TSH levels and low free T4 levels. She is on low-dose aspirin and statins. What would be the appropriate management of this patient?
- a. Start levothyroxine 25µg and escalate based on TSH levels
  - b. Start Levothyroxine 100µg and titrate based on TSH values
  - c. Start Liothyronine 100µg and titrate dose based on TSH values
  - d. Start Liothyronine 25µg and escalate dose based on TSH values

Ans: (a)

- Q. What is the drug of choice to manage thyroid storm?
- a. Methimazole
  - b. Propranolol
  - c. Propylthiouracil
  - d. Hydrocortisone

Ans: (c)



# 12. PANCREATIC NEUROENDOCRINE TUMORS

## INCIDENCE OF PANCREATIC NEUROENDOCRINE TUMORS

00:01:06

- **M/c** functioning pancreatic NET - Insulinoma
- **M/c** non-functioning Pancreatic NET - PP-OMA (Pancreatic polypeptide)
  - Abdominal pain
  - Obstructive jaundice
  - Weight loss
- Cut off for Hypoglycemia: < 54 mg/dL
  - Counter regulatory hormones- Glucagon, Cortisol, Catecholamines & Growth hormone ↑
- T/t of Symptomatic hypoglycemia
  - Oral glucose
  - IV 25-50% Dextrose
  - If IV access unavailable → give Glucagon S/c

|                |                                        |
|----------------|----------------------------------------|
| <b>Level 1</b> | • 55-70 mg %                           |
| <b>Level 2</b> | • <54 mg%                              |
| <b>Level 3</b> | • Any value, requirement of assistance |

## DIFFERENTIAL DIAGNOSIS FOR HYPOGLYCEMIA

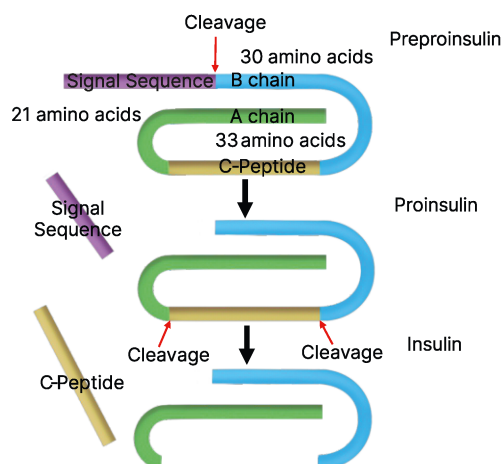
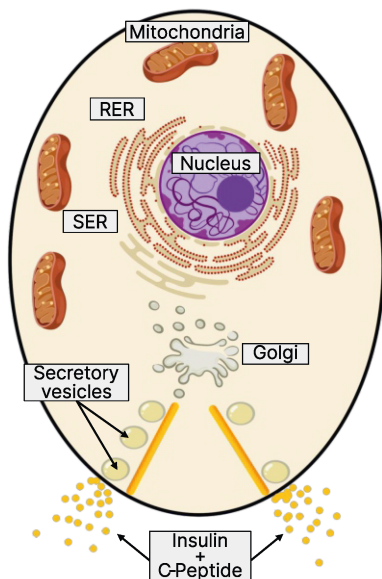
00:07:57

Mnemonic: **CAPE-IAS**

- **C**KD - Insulin is excreted via the kidney
- **C**LD
- **A**lcoholic liver disease
- **P**oor nutrition
- **E**xtra-pancreatic tumor: Leukemia/Lymphoma
- **I**nsulinoma
- **A**buse of **S**ulfonylureas/Insulin

## INSULINOMA

00:12:00





## WHIPPLE'S TRIAD

- Symptoms of hypoglycemia (Emotional outburst)
  - Incidence of malignancy in Insulinoma: 5-15%
  - Usual size at diagnosis → ~2 cm
  - Location of tumor: Head, neck, tail of Pancreas
- Low blood sugar <45 mg/dl
- Immediate relief of symptoms with IV glucose

### Important Information

- More common Pancreatic endocrine tumor in MEN 1 syndrome is ZES > Insulinoma

## CLINICAL FEATURES

00:16:08

- ↑ Frequency of eating and weight gain / insulin ↑
- Emotional lability
- ↓ Concentration span and mood swings
- Palpitations and diaphoresis (Sympathetic stimulation)
- A/w MEN1

## WORK UP

00:18:29

### 72-HOUR FASTING TEST

- Random blood sugar ↓
- RiA-Insulin ↑
- RiA- C-Peptide ↑
- Proinsulin level ↑
- Insulin : Glucagon ratio >0.3 (Normal <0.3)
- Low Plasma β OH butyrate
- To rule out factitious hypoglycemia (done for urine sulfonylureas)

## IMAGING

00:25:51

- Gallium 68 Dotatate PET/CT > Endoscopic ultrasound > Somatostatin receptor Scintigraphy (SRI)

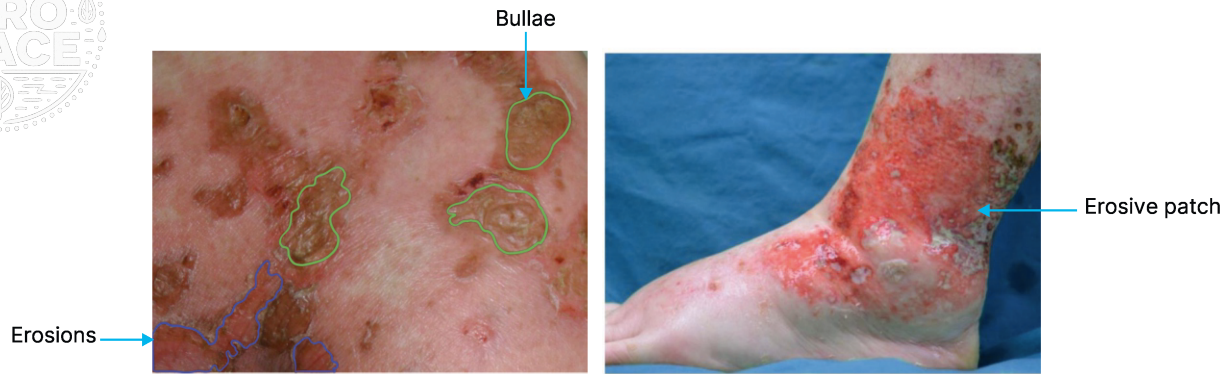
## TREATMENT

00:26:53

1. Frequent small meals
2. DOC: Diazoxide
  - MOA - Open K<sup>+</sup> channels - Hyperpolarization of β cells, Causes → ↓ in release of Insulin
  - Lanreotide - ↓ GH and sugar levels → worsening of symptoms
3. Malignant insulinoma
  - mTOR inhibitor → Everolimus
  - Radiolabelled somatostatin



00:30:12



## CLINICAL FEATURES

00:30:12

- Diabetes mellitus, Migratory necrolytic erythema
- Dermatitis
- Deep vein thrombosis
- Depression

## FEATURES OF GLUCAGONOMA

00:32:07

- Size of the tumor: 5-10 cm
- Location of the tumor - Tail of the pancreas
- Metastasis: ++
- Plasma glucose levels > 1000 pg/mL
- MRI/CT abdomen shows tumor, metastasis +/-

## TREATMENT

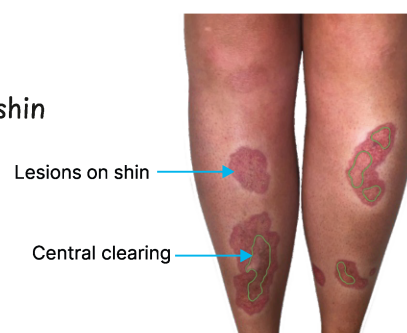
00:33:31

- Radiolabeled Somatostatin analogs
- Debulking surgery

## NECROBIOSIS LIPOIDICA DIABETICORUM

00:33:41

- Lesion (reddish brown with Central clearing) Usually on shin
- Seen in Type 1 DM



## ACANTHOSIS NIGRICANS

00:35:43

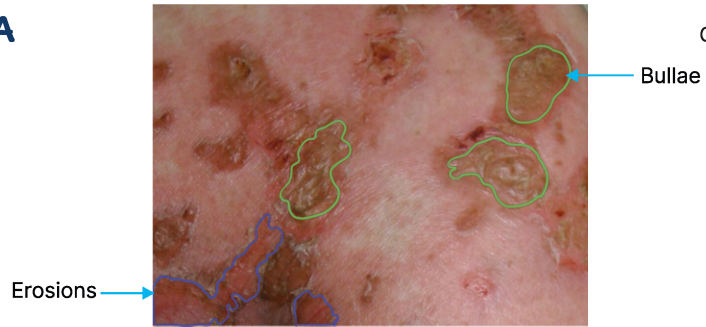
- Seen in Metabolic syndrome
- Velvety dark pigmented plaques on dorsum of neck and axilla



Acanthosis Nigricans

## MIGRATORY NECROLYTIC ERYTHEMA

- **Skin**
  - Glucagonoma
  - Myeloproliferative disorders
  - Malnutrition
  - Short bowel syndrome



00:35:41

Bullae

Erosions

MIGRATORY NECROLYTIC ERYTHEMA

00:37:21

## SOMATOSTATINOMA

### CLINICAL FEATURES

Mnemonic: **DDG**

- **D**iabetes Mellitus: D/t inhibition of insulin
- **D**iarrhea /Steatorrhea: D/t the inhibition of pancreatic amylase
- **G**all bladder sludge/stasis/stones - Inhibition of CCK
- Hypochlorhydria

## VIPOMA (VASOACTIVE INTESTINAL PEPTIDE-SECRETING TUMOR)

00:40:24

- VIP: 28 amino acid peptide
- Secreted by F cells in the pancreas, Carcinoid tumor

### CLINICAL FEATURES

00:41:12

- Pancreatic cholera: **Rice water stool**
- Stool Osmolal gap: < 50-**Secretory diarrhoea**
- Hypokalemia
- Flushing episodes
- Achlorhydria
- **T/t**: Octreotide

### Important Information

- VIPoma → aka **WDHA** Syndrome
  - **W**atery **D**iarrhea
  - **H**ypokalemia
  - **A**chlorhydria

## CAUSES OF SECRETORY DIARRHEA

00:45:27

(Mnemonic - **DM-CL**)

- **D**iabetic Diarrhoea
- **M**edullary carcinoma of thyroid ← Calcitonin, VIP
- **C**arcinoid tumor/Cholera/Pancreatic cholera
- **L**axative abuse

### Important Information

- Stool osmotic gap =  $290 - 2(\text{Na} + \text{K})$ 
  - < 50: Secretory diarrhoea
  - > 100: Osmotic diarrhoea



## PRIMITIVE NEUROECTODERMAL TUMORS ( pNET)

- Example Neuroblastoma

00:48:43

## PERIPHERAL PRIMITIVE NEUROECTODERMAL TUMORS (ppNET)

- Medulloblastoma, Ewing sarcoma

00:49:25



Q. Which neuroendocrine tumor is most likely responsible for recurring episodes of watery diarrhea, dehydration, hypokalemia, and achlorhydria? (NEET PG)

- a. Insulinoma
- b. Glucagonoma
- c. VIPoma
- d. Somatostatinoma

Ans (c)

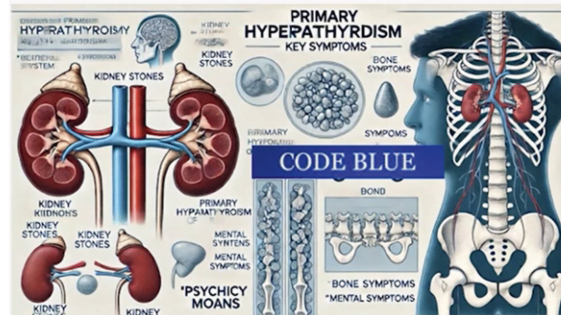
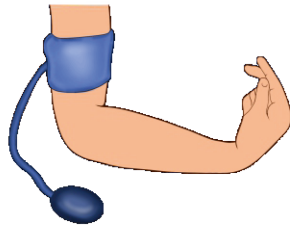
Q. Dermatitis, diarrhea, diabetes mellitus, and deep vein thrombosis are seen with which of the following conditions? (FMGE)

- a. Glucagonoma
- b. Niacin deficiency
- c. Insulinoma
- d. Corn diet

Ans (a)



# 13. DISORDERS OF THE PARATHYROID GLAND



## HORMONES INVOLVED IN MAINTENANCE OF CALCIUM LEVELS

00:01:26

|                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PTH is released from the parathyroid gland and acts on</b>                           | <ul style="list-style-type: none"> <li>Bone: Bone resorption: ↑ Calcium</li> <li>Kidney: Acts on PCT → PO<sub>4</sub> wasting <ul style="list-style-type: none"> <li>↑ Calcium reabsorption in DCT (via CaSR)</li> </ul> </li> <li>Calcium &amp; Phosphate have an <b>inverse relation</b></li> <li>Bone turnover Marker- <b>Serum Alkaline Phosphatase</b></li> <li>Serum Alkaline Phosphatase produces <ul style="list-style-type: none"> <li>Osteoclasts</li> <li>Osteoblasts</li> </ul> </li> </ul> |
| <b>FGF-23 (fibroblast growth factor) is released from bone (osteocytes) and acts on</b> | <ul style="list-style-type: none"> <li>Regulates Phosphate</li> <li>Kidney - Phosphate lowering</li> <li>GIT</li> <li>Parathyroid gland - Inhibits PTH release</li> </ul>                                                                                                                                                                                                                                                                                                                               |
| <b>Calcitonin is released from C cells in the thyroid</b>                               | <b>Limited role in humans</b> <ul style="list-style-type: none"> <li>Inhibit osteoclast-mediated bone resorption: Osteoporosis</li> <li>Promotes calcium clearance in urine - Hypercalcemic crisis</li> </ul>                                                                                                                                                                                                                                                                                           |

## SERUM ALKALINE PHOSPHATASE

00:08:33

- Produced by the Liver, Bone, Placenta

## CHANGES IN SAP(BONE SPECIFIC) VALUES IN VARIOUS MEDICAL CONDITIONS

|                                    |                                                                   |
|------------------------------------|-------------------------------------------------------------------|
| <b>ELEVATED</b>                    | Rickets, Osteomalacia, Hyperparathyroidism - Primary or Secondary |
| <b>DECREASED</b>                   | Hypophosphatasia                                                  |
| <b>NORMAL</b>                      | Hypoparathyroidism, Osteoporosis, Multiple myeloma                |
| <b>DISPROPORTIONATELY ELEVATED</b> | Paget's disease                                                   |